



Visual outcomes of macular melanocytic lesions after early or delayed proton beam therapy

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Abstract

Purpose During their initial management, some macular melanocytic lesions can be closely monitored to wait for a documented growth before advocating a treatment by irradiation. However, the visual outcomes of this strategy have not yet been assessed. This study compares the visual outcomes of macular melanocytic lesions that underwent delayed proton beam therapy (PBT) after an initial observation to those treated early.

Methods A total of 162 patients with suspicious melanocytic lesions whose margins were located within 3 mm of the fovea were recruited from two French ocular oncology centers.

Results Overall, 82 patients treated with PBT within 4 months after the initial visit (early PBT group) were compared to 24 patients treated with delayed PBT (delayed PBT group) and 56 patients not treated with PBT (observation group). Visual acuity was not significantly different between baseline and last visit in the observation group ($p=0.325$). Between baseline and last visit, the median [IQR] loss in visual acuity was significant in both the early (0.7 [0.2; 1.8], $p<0.001$) and the delayed (0.5 [0.2; 1.5], $p<0.001$) PBT groups. After irradiation, there was no significant difference between the early and delayed PBT groups for visual loss ($p=0.575$), diameter reduction ($p=0.190$), and thickness lowering ($p=0.892$). In multivariate analysis, history of diabetes mellitus and Bruch's membrane rupture remained significantly associated with greater visual loss ($p=0.036$ and $p=0.002$, respectively).

Conclusion For small lesions in which there is no clear diagnosis of malignant melanoma, an initial close monitoring to document tumor growth does not impact visual prognosis, despite the potential complications associated with the untreated tumor. However, the survival should remain the main outcome of the treatment of these lesions.

Keywords Macular lesion · Proton beam therapy · Radiation maculopathy · Uveal melanoma

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Key messages

- Proton beam therapy (PBT) in macular melanocytic tumors is associated with significant loss of vision.
- For small suspicious melanocytic lesions located within 3 mm of the fovea with no clear diagnosis of malignant melanoma, an initial close monitoring to document tumor growth does not impact visual prognosis, despite possible compressive or exudative complication.

Introduction

Uveal melanoma (UM) is the most frequent primary intraocular malignancy and can be associated with death in case of systemic dissemination [1]. Historically, enucleation was the gold standard treatment, but conservative treatments have since proven their efficacy and show similar survival rates and no increased risk of metastasis [2–4]. The two major radiotherapy methods currently used are proton beam therapy (PBT) and plaque brachytherapy. Both techniques allow a precise delivery of the radiation dose, thus sparing the healthy surrounding tissue from useless irradiation [1]. These 2 treatment modalities have shown effective outcomes but, as a charged-particle therapy, PBT can provide highly localized radiation dose distribution with sharp distal and lateral margins. However, despite delivering a precise irradiation of the tumor, PBT can be associated with several complications [5–7]. Therefore, it is important to treat only highly suspicious melanocytic lesions at risk of UM. The diagnosis of such lesions however, which often present as flat and narrow, can rarely be performed upon biopsy. To this aim, some authors have described risk factors of tumor growth defining a diagnosis of UM using clinical examination and multimodal imaging [8]. When the diagnosis of malignant melanoma is not clear, it has been shown that a close monitoring of the lesion to document tumor growth before treatment does not affect metastatic and survival rates [9]. However, little is known about the visual prognosis in this wait-and-see policy, an issue which needs to be addressed as some melanocytic lesions may present complications such as exudation or neovascularization, resulting in decreased vision [10]. This is particularly true for macular lesions for which complications can directly affect the macula, but for which irradiation puts the patient at high-risk of vision loss by inducing radiation retinopathy [11, 12]. One can therefore hypothesize that delaying treatment of macular tumors may induce worse visual outcome by allowing tumor growth, thus decreasing the distance to

the macula and increasing the risk of compressive and exudative complications.

The aim of this study was to compare the visual outcomes of macular melanocytic lesions that underwent delayed PBT after an initial observation to those treated early. The secondary objective was to characterize predictive factors of worse visual loss after irradiation.

Materials and methods

This retrospective study included consecutive patients with a melanocytic lesion localized in the macular area, who were diagnosed from 2009 onwards, and who were followed between January, 2019 and June, 2021 in two French reference ocular oncology centers: *Rétine Gallien* (Bordeaux) and *Hopital de la Croix-Rousse* (Lyon). The inclusion criterion was a diagnosis of a choroidal nevus or a choroidal melanoma whose margin was within 3 mm of the fovea. Only patients with at least one risk factor of tumor growth (thickness > 2 mm, presence of subretinal fluid (SRF), symptoms of visual acuity loss to 20/50 or worse, presence of orange pigment, presence of tumor acoustic hollowness in ultrasonography, and diameter > 5 mm, as defined by Shields et al. [8]) were included. Exclusion criteria were a follow-up of less than 12 months and a history of previous tumor treatment before referral. Research was conducted in accordance with the Declaration of Helsinki. An international review board approved the study (Ethics Committee of the French Society of Ophthalmology, IRB 00008855 Société Française d'Ophtalmologie IRB#1).

Data collection

Data retrieved at baseline were age, sex, phakic status, history of previous oncologic disease, presence of high blood pressure and/or diabetes mellitus, and best corrected visual acuity (BCVA). Moreover, the following tumor characteristics were recorded: initial thickness and diameter, distance to the fovea and the optic disc, presence of SRF, orange pigments,

depigmented halo, Bruch's membrane rupture, extrascleral extension, optic nerve invasion, ultrasonographic hollowness, and visual symptoms. The presence of systemic metastasis was also assessed using full body computed tomography (CT).

At each follow-up visit, BCVA was measured, and possible complications were assessed. Slit-lamp examination, fundoscopy, and multimodal imaging results including spectral-domain optical coherence tomography (SD-OCT), retinography, fluorescein angiography, indocyanine green angiography, and ultrasonography were used to assess the local tumoral control and the presence of complications. After treatment, complications of radiation maculopathy were assessed by SD-OCT for the presence of macular edema, at each visit. The presence of ischemia was assessed through a combination of fundoscopy, ultra-wide field retinography, OCT-angiography, and/or fluorescein angiography, at the discretion of the ophthalmologist.

Systemic metastasis was systematically assessed using liver ultrasonography or magnetic resonance imaging (MRI) each 3 months during the first two years and each 6 months thereafter. Further CT or MRI were used for suspicious cases.

Data concerning other treatments prescribed for specific tumor exudation or radiation-induced retinopathy/maculopathy were also collected: photodynamic therapy (PDT), laser photocoagulation, intravitreal injections (IVT) of anti-vascular endothelial growth factor (anti-VEGF: aflibercept, ranibizumab, or bevacizumab), or dexamethasone implant, and enucleation.

Groups of patients

Treatment with PBT was decided for tumors with a high suspicion of malignancy as well as tumors with numerous growth risk factors or with documented growth during follow-up.

Then, the patients were classified into 3 groups according to their treatment: the “early PBT group” included patients who underwent PBT within the 4 months following baseline assessment; the “delayed PBT group” included patients who underwent PBT later than 4 months after the initial visit; and the “observation group” included patients who were managed without irradiation. For the latter, no local tumor growth was observed during follow up (Fig. 1).

Proton beam therapy

For patients treated with PBT, 4 tantalum fiducials (clips) were positioned on the sclera under anesthesia using transillumination. Clip to clip, tumor to clips, tumor to macula, and optic disc distances were reported. All patients were treated at the cyclotron of the *Centre Lacassagne* in Nice (France). Irradiated patients underwent a 60_Gray Relative Biological Effectiveness (Gy RBE) hypofractionated irradiation delivered over the course of 4 days. Gaze fixation was optimized for preservation of ocular structures. A 2.5-mm margin was set by adjusting the range and modulation width of the spread-out Bragg peak (SOBP) plateau's proximal and distal edges. Beam modifiers were designed depending of the

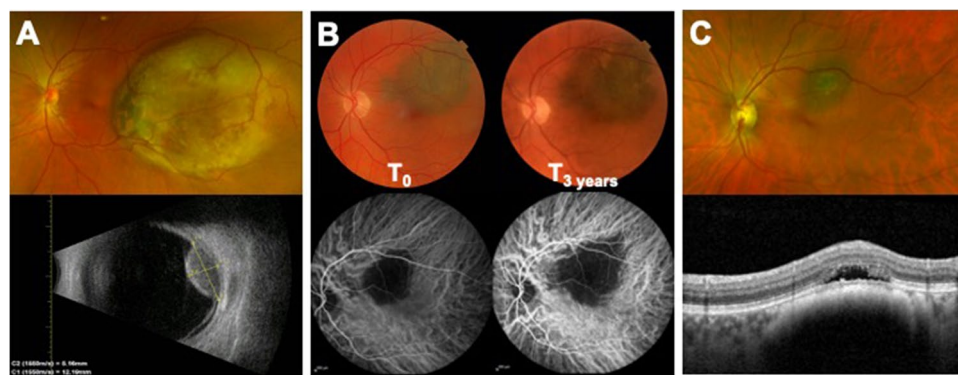


Fig. 1 Illustrating cases of patient in each group. **A** Early proton beam therapy (PBT) group. Retinography showed large and thick amelanotic tumor surrounded by subretinal fluid and covered by orange pigments. Ultrasonography found retinal detachment and choroidal tumor with high echogenicity on its surface and low echogenicity inside. **B** Delayed PBT group. Retinography showed suspicious

pigmented macular lesion with evolution over 3 years. The marginal extension was more evident on indocyanine green angiography. **C** Observation group. Retinography showed small and flat pigmented tumor with subretinal fluid. No growth was observed during follow-up

target range, on the hypothesis of a rectilinear trajectory of protons. Lateral conformation was achieved by a brass collimator adjusted to the tumor contours. The eye position was monitored by video camera during the 10-s treatment, eye movements triggering immediate interruption of the beam.

Outcomes

The primary outcome was the BCVA recorded at last visit obtained using Snellen's scale and converted into LogMAR. Patients who had only light perception were assigned the value of 2.7 LogMAR and those with no light perception were assigned the value 3.0 LogMAR. Secondary outcomes were the presence of ocular complications such as cataract, ocular hypertension, inflammation, vitreous hemorrhage, dry eye syndrome, radiation retinopathy, radiation papillopathy, and neovascular glaucoma. Local recurrence and systemic metastasis were also recorded.

Statistical analysis

Quantitative parameters were described by mean and standard deviation (SD) or by median and interquartile range [IQR] according to the normality of the distribution assessed by the Shapiro–Wilk test. Qualitative parameters were expressed as frequency and percentage. Patient and tumor characteristics, and treatments and complication rates were compared using a chi-square test or Fisher's exact test for qualitative parameters and one-way ANOVA test or Kruskal–Wallis test for quantitative parameters. Significant loss of visual acuity within each group was assessed using a paired Wilcoxon signed-rank test. All analyses for comparison of visual acuity at last follow-up between the 3 groups were adjusted on follow-up duration and BCVA at diagnosis. Consequently, the quantitative LogMAR values were compared in a multivariate linear regression model, and the ordinal values obtained using Snellen's classification and the loss of ETDRS lines were compared by multivariate ordinal logistic regression analysis. The proportion of patients who lost ≥ 1 line was compared between each group by multivariate logistic regression analysis. The same models were applied to investigate the visual outcomes according to the presence of radiation maculopathy. The change between before and after PBT decision in visual acuity, tumor diameter, and tumor thickness was investigated within each group using a paired Wilcoxon signed-rank test or a paired *t*-test. The comparison of values between early and delayed PBT groups was performed by a Mann–Whitney U test or Student *t*-test at PBT decision and with the previous described multivariate linear regression at last follow-up. Predictive factors of worse visual acuity after PBT were first tested in bivariate analyses by adjusting on visual acuity just before PBT time with multivariate linear regression; parameters

with a *p*-value under 0.1 were then introduced in a full multivariate linear regression and this full model was simplified by a backward selection in line with the parsimony principle. Results were presented as estimated coefficients and their standard deviation. For all linear regression analyses, the model assumptions (as the normality and independence of residuals error terms) and the impact of outliers were investigated. Statistical analyses were performed using SAS software, version 9.4 (SAS Institute Inc., Cary, NC, USA). Significant level was set at 0.05.

Results

Patient characteristics

Overall, 183 patients were screened in this study. Among these patients, 162 were included because of suspected malignancy due to the presence of one or more predictive criteria of malignancy. Among them, 82 patients (50.6%) were treated with PBT in the first 4 months following baseline assessment (median [IQR]: 1.5 months [1; 2]), 24 patients (14.8%) were treated with PBT more than 4 months after initial visit (median [IQR]: 17.5 months [10.5; 57]), and 56 patients (34.6%) were not treated with PBT. The main characteristics of patients at baseline were not significantly different between the 3 groups except for median BCVA and history of high blood pressure (Table 1).

Tumor characteristics

The mean (SD) distance to fovea was 1.3 (1.0) mm for the total cohort. The following tumor characteristics at baseline were significantly different between the 3 groups: mean diameter, mean thickness, presence of amelanotic lesion, presence of a Bruch's membrane rupture, the number of malignancy predictors, the initial diagnosis made, and all predictors of malignancy except the presence of orange pigments (Table 1).

Time of PBT treatment

In the delayed PBT group, decision to treat with PBT was made because of an increase in mean (SD) tumor diameter (+1.6 (1.6) mm, $p < 0.01$), a rise in mean (SD) thickness (+0.89 (1.4), $p < 0.01$), or the appearance of ultrasonographic hollowness ($p = 0.008$). At the time of PBT treatment, early and delayed PBT groups were significantly different in terms of median [IQR] visual acuity (0.3 [0.1;0.5] LogMAR versus 0 [−0.1;0] LogMAR, respectively, $p < 0.001$), mean (SD) tumor diameter (10.5 (2.8) mm versus 8.1 (1.4) mm), respectively, $p < 0.001$), and mean (SD)

Table 1 Patient and tumor characteristics at baseline according to management strategy

	Early PBT group N=82	Delayed PBT group N=24	Observation group N=56	p-value
Male, n (%)	39 (47.6)	11 (45.8)	22 (39.3)	0.623
Mean age, years (SD)	65 (16)	66 (16)	65 (14)	0.990
Lens status, phakic, n (%)	69 (84.2)	21 (87.5)	44 (78.6)	0.556
History of diabetes mellitus, n (%)	8 (9.8)	1 (4.2)	3 (5.4)	0.646
History of high blood pressure, n (%)	20 (24.4)	13 (54.2)	10 (17.9)	0.003
Previous oncologic medical history, n (%)	18 (22)	3 (12.5)	8 (14.3)	0.389
Median BCVA, LogMAR [IQR]	0.3 [0.1;0.5]	0.02 [0;0.2]	0.05 [0;0.2]	<.001
Mean distance to fovea, mm (SD)	1.4 (1.0)	1.5 (1.2)	1.2 (1.0)	0.438
Mean distance to optic disc, mm (SD)	3.3 (1.9)	2.7 (2.1)	2.8 (1.9)	0.301
Mean diameter, mm (SD)	10.5 (2.8)	6.5 (1.6)	6.2 (2.7)	<.001
Mean thickness, mm (SD)	4.9 (2.3)	2.2 (1.1)	1.5 (1.0)	<.001
Tumoral color, white, n (%)	17 (20.7)	5 (20.8)	3 (5.4)	0.036
Bruch's membrane rupture, n (%)	9 (11)	1 (4.2)	0 (0)	0.021
Extrascleral extension, n (%)	1 (1.2)	0 (0)	0 (0)	NC
Predictors of malignancy*				
Thickness > 2 mm, n (%)	79 (96.3)	12 (50)	10 (17.9)	<.001
Peritumoral SRF, n (%)	69 (85.2)	17 (70.8)	26 (46.4)	<.001
Visual acuity loss, n (%)	73 (89)	14 (58.3)	23 (41.1)	<.001
Orange pigments, n (%)	36 (44.4)	11 (45.8)	21 (37.5)	0.669
Ultrasonographic hollowness, n (%)	77 (93.9)	16 (66.7)	4 (7.1)	<.001
Diameter > 5 mm, n (%)	79 (96.3)	21 (87.5)	38 (67.9)	<.001
Number of malignancy predictors, n (%)*				<.001
1	0 (0)	3 (12.5)	20 (35.7)	
2	1 (1.2)	3 (12.5)	20 (35.7)	
3	4 (4.9)	3 (12.5)	6 (10.7)	
4	11 (13.4)	6 (25)	7 (12.5)	
5	41 (50)	5 (20.8)	2 (3.6)	
6	25 (30.5)	4 (16.7)	1 (1.8)	
Other predictors				
< 3 mm to disc, n (%)	33 (40.7)	11 (45.8)	27 (48.2)	0.675
Absence of depigmented halo, n (%)	80 (98.8)	22 (91.7)	51 (91.1)	0.063
Initial diagnostic, n (%)				<.001
Atypical naevus	1 (1.2)	19 (79.2)	56 (100)	
Melanoma	81 (98.8)	5 (20.8)	0 (0)	

BCVA, best-corrected visual acuity; IQR, interquartile range; PBT, proton beam therapy; SD, standard deviation; SRF, subretinal fluid; NC, not computed

* As defined by Shields et al. [8]

tumor thickness (4.9 (2.3) mm versus 3.1 (1.3) mm, respectively, $p < 0.001$).

Functional outcome

The median [IQR] duration of follow-up was 33 months [23;56] after the initial visit in the observation group, 42 months [26;63] after PBT in the early PBT group, and 47 months [29;65] after PBT in the delayed PBT group ($p = 0.358$). At the end of follow-up, loss of visual acuity

was significant in the early PBT and delayed PBT groups with a median [IQR] loss of 0.7 [0.2;1.8] ($p < 0.001$) and 0.5 [0.2; 1.5] ($p < 0.001$), respectively. Visual acuity was not significantly different between baseline and last visit in the observation group ($p = 0.325$). Median [IQR] BCVA at last follow-up was 1.3 [0.5; 2.3] LogMAR in the early PBT group, 0.7 [0.2; 1.7] LogMAR in the delayed PBT group, and 0.05 [0; 0.2] LogMAR in the observation group ($p < 0.001$). There was a significant difference between the 3 groups for the loss of ETDRS lines ($p < 0.001$), and the

Table 2 Variations in visual acuity during follow-up

	Early PBT group <i>N</i> = 82	Delayed PBT group <i>N</i> = 24	Observation group <i>N</i> = 56	<i>p</i> -value ^a
Median visual acuity at final follow-up, LogMAR [IQR]	1.3 [0.5;2.3]	0.7 [0.2;1.7]	0.05 [0;0.2]	< .001
Median difference of visual acuity since baseline, LogMAR [IQR]	0.7 [0.2;1.8]	0.5 [0.2;1.5]	0 [-0.07;0]	
Final visual acuity, <i>n</i> (%)				
≥ 20/40	15 (18.5)	8 (33.3)	51 (91.1)	< .001
20/50–20/200	22 (27.2)	7 (29.2)	5 (8.9)	
< 20/200	34 (42)	6 (25)	0 (0)	
No LP	10 (12.4)	3 (12.5)	0 (0)	
Loss of ETDRS lines, <i>n</i> (%)				
0	17 (21.5)	5 (20.8)	50 (89.3)	< .001
1 line	4 (5.1)	2 (8.3)	3 (5.4)	
2 lines	3 (3.8)	2 (8.3)	1 (1.8)	
≥ 3 lines	55 (69.6)	15 (62.5)	2 (3.6)	
Proportion of patients with EDTRS line loss, <i>n</i> (%)				
≥ 1 line	62 (78.5)	19 (79.2)	6 (10.7)	< .001
≥ 2 lines	58 (73.4)	17 (70.8)	3 (5.4)	< .001
≥ 3 lines	55 (69.6)	15 (62.5)	2 (3.6)	< .001

IQR, interquartile range; *LP*, light perception, *PBT*, proton beam therapy

^a Adjusted on duration of follow-up and BCVA at diagnosis

proportion of patients who lost ≥ 1 line ($p < 0.001$), ≥ 2 lines ($p < 0.001$), and ≥ 3 lines ($p < 0.001$; Table 2).

Comparison of outcomes according to PBT group

Median [IQR] time before irradiation was 1.6 months [1.3;2] and 17.5 months [10.5;57] in the early PBT and delayed PBT groups, respectively.

After irradiation, there was no significant difference between early PBT and delayed PBT groups for visual acuity loss ($p = 0.575$), diameter reduction ($p = 0.190$), and thickness lowering ($p = 0.892$; Table 3).

Predictive factors of worse visual acuity after irradiation

Table 4 presents the bivariate analysis for predictive factors of worse visual acuity following PBT, after adjustment on visual acuity just before PBT decision. In the multivariate analysis, history of diabetes mellitus and Bruch's membrane rupture remained significantly associated with a greater severity of visual loss (estimate (SD) 0.674 (0.316), $p = 0.038$ and 0.966 (0.302), $p = 0.002$, respectively).

Complications

Overall, 4 patients (4.8%) in the early PBT group, 1 (4.1%) in the delayed PBT group, and 0 in the observation group underwent secondary enucleation. In the early PBT group, enucleation was performed because of local relapse in 50.0% of cases, neovascular glaucoma in 25.0%, and chronic total retinal detachment in 25.0%. Liver metastases appeared during follow-up in 8.0% of cases in the early PBT group, and none occurred in the delayed PBT and observation groups ($p = 0.194$). No other systemic metastasis localization was found. Radiation maculopathy appeared in 50 patients (61.0%) and 14 patients (56.0%) in the early PBT and delayed PBT groups, respectively ($p = 0.324$). Presence of both ischemia and macula edema was more commonly noted, with 29 patients (35.4%) and 6 patients (25.0%) in the early PBT and the delayed PBT groups, respectively (Table 5). At the end of follow-up, visual acuity was not significantly different between patients with radiation maculopathy (1.3 [0.49; 2.3]) and those without radiation maculopathy (0.65 [0.22; 2.3], $p = 0.454$). According to the presence or absence of radiation maculopathy, there was a significant difference for loss of ETDRS lines ($p = 0.047$), and the proportion of patients who lost ≥ 1 line ($p = 0.047$), ≥ 2 lines ($p = 0.042$), but not for ≥ 3 lines ($p = 0.070$; Table 6).

Table 3 Changes in visual acuity, and tumor diameter and thickness after irradiation

	Early PBT group N = 82	Delayed PBT group N = 24	p-value
Median visual acuity, LogMAR [IQR]			
At PBT decision	0.3 [0.1;0.5]	0 [-0.1;0]	< .001 ^b
At last visit	1.3 [0.5;2.3]	0.7 [0.2;1.7]	0.575 ^c
Difference	0.7 [0.2;1.8]	0.7 [0.2;1.6]	
p	< .001 ^a	< .001 ^a	
Mean diameter, mm (SD)			
At PBT decision	10.5 (2.8)	8.1 (1.4)	< .001 ^b
At last visit	8.4 (3.6)	7.2 (1.7)	0.190 ^c
Difference	-2.2 (3.0)	-0.8 (1.8)	
p	< .001 ^a	0.044 ^a	
Mean thickness, mm (SD)			
At PBT decision	4.9 (2.3)	3.1 (1.3)	< .001 ^b
At last visit	3.0 (2.2)	1.9 (0.8)	0.892 ^c
Difference	-1.7 (2.0)	-0.9 (1.0)	
p	< .001 ^a	< .001 ^a	

IQR, interquartile range; PBT, proton beam therapy; SD, standard deviation

^a Evolution between PBT decision and last visit

^b Comparison between early and delayed PBT groups for visual acuity, diameter, and thickness at PBT decision

^c Comparison between early and delayed PBT groups for changes in visual acuity, diameter, and thickness after PBT adjusted on values just before PBT time and duration of follow-up after PBT

Other treatment

The PDT were all performed for exudative complications. Focal laser was used for polypoidal vasculopathy in 10 patients (91.0%) and radiation maculopathy in 1 patient (9.0%). In the case of retinal ischemia appearance, retinal photocoagulation was performed to prevent neovascular complications. Intravitreal injection of anti-VEGF was administered for radiation maculopathy in 36 patients (60.0%), neovascular glaucoma in 11 patients (17.5%), toxic tumor syndrome in 6 patients (9.5%), and polypoidal vasculopathy or macular neovascularization in 8 patients (13.0%). Ranibizumab was used in 30 patients (49.0%), aflibercept in 17 patients (28.0%), and bevacizumab in 14 patients (23.0%). Intravitreal injection of dexamethasone implant was performed for radiation maculopathy in 17 patients (50.0%), toxic tumor syndrome in 10 patients (29.0%), exudative retinal detachment in 6 patients (18.0%), and central venous occlusion in 1 patient (3.0%). The use of these treatments was significantly different between the 3 groups (Table 7).

Discussion

This study presents a cohort of patients with macular melanocytic lesions located close to the fovea. The results herein indicate that PBT treatment of choroidal melanoma located within 3 mm of the fovea is strongly associated with a decrease in visual acuity. Delaying PBT to observe tumor

Table 4 Bivariate analysis of predictive factors of worse visual acuity following PBT, after adjustment on visual acuity just before PBT decision

	Estimate (SD)	p-value ^c
Male	0.309 (0.187)	0.101
Age, years	0.007 (0.006)	0.202
Lens status, phakic	-0.106 (0.263)	0.686
History of diabetes mellitus	0.696 (0.33)	0.038
History of high blood pressure	0.115 (0.204)	0.573
History of previous systemic oncologic	-0.182 (0.235)	0.440
Distance to fovea ^a , mm	-0.003 (0.098)	0.972
Distance to optic disc ^a , mm	-0.096 (0.049)	0.052
Diameter ^b , mm	0.007 (0.035)	0.838
Thickness ^b , mm	0.132 (0.046)	0.005
Tumoral color, white	-0.05 (0.233)	0.832
Bruch's membrane rupture	0.98 (0.308)	0.002
Predictors of malignancy ^{b*}		
Thickness > 2 mm	0.045 (0.408)	0.913
Peritumoral SRF	0.081 (0.252)	0.749
Visual acuity loss	0.522 (0.268)	0.055
Orange pigments	0.027 (0.185)	0.886
Ultrasonographic hollowness	0.381 (0.405)	0.349
Diameter > 5 mm	0.559 (0.564)	0.325
Number of malignancy predictors ^b	0.192 (0.107)	0.075
Other predictors ^b		
< 3 mm to disc	0.208 (0.187)	0.267
Absence of depigmented halo	0.344 (0.687)	0.618
Melanoma at diagnosis	0.013 (0.245)	0.957

SD, standard deviation; SRF, subretinal fluid

^a At diagnosis

^b Just before PBT time

^c Adjusted on visual acuity just before PBT time

growth, however, was not associated with worse visual prognosis compared to early PBT treatment. Over a follow-up period of 3–4 years, patients who were not irradiated showed no significant variation in visual acuity. It should be noted however, that these patients had significantly more favorable tumor characteristics (thinner, smaller, less exudative), even if they had to have at least 1 criterion of malignancy to be included. Nevertheless, this observation group was a useful comparator for the treatment groups and allowed to analyze the possible non-malignant complications of these lesions.

More than half of the patients who underwent PBT in the present study maintained a useful vision defined by a visual acuity > 20/200. Few studies have reported visual outcome in patients with macular choroidal melanoma treated with irradiation. The Collaborative Ocular Melanoma Study (COMS) report n°16 also found that proximity between the tumor and the foveal avascular zone was associated with worse visual outcome at 3 years following plaque brachytherapy [13]. Gunduz et al. reported the outcome of 630 patients with choroidal melanoma involving the macula (margin within 3 mm of the fovea) and treated with plaque brachytherapy [14]. In their study, 40% of patients lost at least 3 lines of visual acuity 5 years after treatment. They found that visual acuity was more than 20/200 in 44% of patients and better than 20/40 in 9% of patients which is slightly less than in the present study. Another cohort of 50 subfoveal melanomas treated with plaque brachytherapy demonstrated that 50% of patients maintained visual acuity > 20/200 [15]. Concerning

PBT, a series of 351 patients with UM located within 1 optic disc diameter from the fovea showed that more than a third (35.5%) retained 20/200 or better vision 5 years after PBT [16].

The present study found that radiation maculopathy was associated with patients who lost vision after PBT, confirming results of previous studies showing that radiation maculopathy is the main contributor to vision loss in patients with tumors located close to the fovea; it has even been shown that the proximity of the tumor to the fovea was the main risk factor for vision loss [11, 17, 18]. Herein, approximately two thirds of the patients irradiated developed radiation maculopathy during follow-up. However, for tumors located in a circle of 3 mm from the fovea, there was no significant difference in final visual acuity according to fovea-tumor distance. This is in line with another study that also did not show a significant difference in the development of radiation maculopathy between tumors located within 1 mm of the fovea and those at a greater distance [19]. This can be explained by the use of 2.5 mm lateral margins during PBT which does not allow to spare the fovea completely. Plan optimization is therefore necessary to deliver a dose as low as reasonably achievable to the fovea. To this aim, new treatment planning protocols have been developed such as hypofractionated PBT [20], or the reduction of lateral margins to 1 mm despite the report of a higher rate of local recurrence [21]. In the present study, results also showed that the presence of diabetes mellitus and the presence of a Bruch's

Table 5 Complication and enucleation rates according to management groups

	Early PBT group N = 82	Delayed PBT group N = 24	p-value ^a	Observation group N = 56
Dry eye	23 (28.1)	6 (25)	0.768	0 (0)
Cataract	67 (81.7)	15 (62.5)	0.048	6 (10.7)
Ocular hypertonia	17 (20.7)	5 (20.8)	1	0 (0)
Vitreous hemorrhage	13 (15.9)	3 (12.5)	1	0 (0)
Intraocular inflammation	5 (6.1)	3 (12.5)	0.377	0 (0)
Radiation maculopathy			0.324	
Ischemia	12 (14.6)	2 (8.3)		
Edema	9 (11)	6 (25)		-
Both	29 (35.4)	6 (25)		
Radiation retinopathy	44 (53.7)	11 (45.8)	0.500	-
Radiation papillopathy	6 (7.3)	2 (8.3)	1	-
Neovascular glaucoma	10 (12.2)	4 (16.7)	0.516	0 (0)
Tumor recurrence	2 (2.4)	0 (0)	NC	
Metastasis	8 (9.8)	0 (0)	0.194	0 (0)
Enucleation	4/82 (4.8)	1/24 (4.1)	1	-
Tumor recurrence	2	0		-
Neovascular glaucoma	1	1		-
Total retinal detachment	1	0		-

Results presented as n (%)

PBT, proton beam therapy; NC, not computed

^a Comparison between early and delayed PBT groups

Table 6 Visual outcomes according to the presence of radiation maculopathy

	Radiation maculopathy		<i>p</i> -value ^a
	No <i>N</i> =42	Yes <i>N</i> =64	
Median visual acuity at final follow-up, Log-MAR [IQR]	0.65 [0.22; 2.3]	1.3 [0.49; 2.3]	0.454
Final visual acuity, <i>n</i> (%)			0.026
≥ 20/40	14 (33.3)	9 (14.3)	
20/50–20/200	11 (26.2)	18 (28.6)	
< 20/200	10 (23.8)	30 (47.6)	
No LP	7 (16.7)	6 (9.5)	
Loss of ETDRS lines, <i>n</i> (%)			0.047
0	13 (31.7)	9 (14.5)	
1 line	3 (7.3)	3 (4.8)	
2 lines	2 (4.9)	3 (4.8)	
≥ 3 lines	23 (56.1)	47 (75.8)	
Proportion of patients who lost			
≥ 1 line	28 (68.3)	53 (85.4)	0.047
≥ 2 lines	25 (61.0)	50 (80.6)	0.042
≥ 3 lines	23 (56.1)	47 (75.8)	0.070

IQR, interquartile range, *LP*, light perception

^a Adjusted on duration of follow-up and BCVA at diagnosis

membrane rupture were associated with worse visual acuity after PBT. Although the diabetes disease, which is independent of the tumor, is a well-known marker of poor functional prognosis [18, 22], a Bruch's membrane rupture has only been reported as a predictive factor of complete vision loss after Gamma Knife stereotactic radiosurgery for UM [23]. Surprisingly and in contrast with other studies [16, 24], the distance to the optic disc was not found herein to be an independent factor of vision loss. This could be explained by the low rate of radiation-induced optic neuropathy in the present study. Although the occurrence of a radiation maculopathy and/or a radiation-induced neuropathy is associated with poor visual outcomes, the early detection of systemic secondary lesions remains a major challenge during follow-up. Frequent and regular liver imaging by ultrasonography or

MRI should be prescribed in order to prompt specific oncologic management when a lesion is detected [25].

As suggested by Vigues-Jorba et al., delayed irradiation could be safely proposed for small choroidal lesions when diagnosis of malignant melanoma is not clear. The authors recently studied a cohort of 75 patients who were diagnosed with small choroidal melanocytic lesions (i.e., thickness < 5 mm and diameter < 10 mm) with growth risk factors. These were treated with brachytherapy after being initially observed and showing progression during follow-up [9]. Five of these patients (6.6%) developed local recurrence requiring enucleation and 4 patients (5.3%) developed metastasis. Survival was 98% at 5 years and 91.6% at 10 and 15 years, confirming a similar prognosis to that of tumors conventionally treated with early irradiation at diagnosis [2, 26, 27]. Some studies have investigated the doubling time of UM and hypothesized that metastatic spread occurs years before the detection of the cancer [28, 29]. Moreover, intrinsic tumor features such as cytomorphology and gene expression have been shown to be the strongest predictors of survival in UM patients [30]. Considering the studies suggesting that metastatic spread occurs early during the development of UM, many years before the diagnosis, and that genetic modifications drive the metastatic risk, it can be hypothesized that waiting for a documented growth of the lesion before irradiating may not lead to an increased risk of systemic metastasis. In the present work, which focused on the visual prognosis of patients who were monitored closely for suspected melanocytic lesions in order to detect early a documented growth before treating, there was no significant difference in loss of visual acuity among patients treated early and those who were initially monitored. Moreover, the reduction in thickness and diameter was not significantly different between the early and delayed PBT groups. Herein, the decision to finally apply tumor irradiation was due to lesion growth. Since these small suspicious tumors were closely monitored to avoid missing tumor growth, it makes sense that at the time of PBT treatment, there was still a difference in visual acuity as well as tumor diameter and thickness

Table 7 Other treatments than irradiation performed

	Early PBT group <i>N</i> =82	Delayed PBT group <i>N</i> =24	Observation group <i>N</i> =56	<i>p</i>
Photodynamic therapy	0 (0)	0 (0)	9 (16.1)	<.001
Focal laser	1 (1.2)	3 (12.5)	7 (12.5)	0.007
Pan retinal photocoagulation	36 (43.9)	9 (37.5)	0 (0)	<.001
Intravitreal injections of anti-VEGF	38 (46.3)	12 (50)	11 (19.6)	0.003
Intravitreal injections of dexamethasone	26 (31.7)	6 (25)	2 (3.6)	<.001

PBT, proton beam therapy; *VEGF*, vascular endothelial growth factor

between the early and delayed PBT groups. However, other ocular complications, such as tumor exudation and choriocapillaris compression, may occur during the monitoring period, thus worsening the visual prognosis in these patients compared to those treated early. The present results suggest that the close monitoring of small suspicious macular tumors does not lead to worse final visual acuity when there is a close monitoring of tumor growth that can prompt PBT decision. These results are thus in accordance with the current paradigm which is characterized by a wait-and-see approach whereby, in the absence of malignant histological proof, patients with small suspicious tumors are not treated until there is a documented growth (i.e., malignant tumor proliferation) [31–33].

Several limitations pertaining to the present study should be acknowledged. Firstly, the comparison between the groups may be challenging as these did not share similar baseline characteristics. This is due to the inclusion criteria which entailed that in the early PBT group patients had a more advanced disease. To overcome this issue, we performed rigorous statistical analyses using a multivariate linear regression model. However, these differences in tumor baseline characteristics may still influence the results. Secondly, despite a long follow-up of almost 4 years in irradiated patients, it is possible that some complications occurred later. It has been shown however, that radiation maculopathy, which is the main cause of irreversible visual loss, generally occurs around 18–24 months after macular irradiation [34]. The retrospective design of the study did not allow to establish the cause of visual decrease in patients irradiated, since these can be various and tumor- or irradiation-dependent. Moreover, even though the measure of visual acuity is intrinsically linked with radiation maculopathy, the assessment of this complication may have been underestimated, as fluorescein angiography and OCT-angiography were not performed at each visit. It should also be noted that the absence of consensus concerning the diagnosis and classification of radiation maculopathy may have further over- or underestimated its incidence. Finally, this study was designed to analyze visual prognosis and complications. The relatively short follow-up duration and the number of patients included in each group prevented the survival rate from being assessed.

The present study found that waiting for a documented growth in melanocytic lesions located within 3 mm of the fovea is not associated with worse visual outcomes, despite potential exudative or compressive complications. For suspicious melanocytic lesions that do not fulfill all uveal melanoma criteria, delaying treatment until tumor growth is documented could be envisioned without any additional risk in terms of visual loss.

Declarations

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the Ethics Committee of the French Society of Ophthalmology, (IRB 00008855 Société Française d’Ophtalmologie IRB#1) and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all individual participants included in the study.

Conflict of interest The authors declare no competing interests.

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